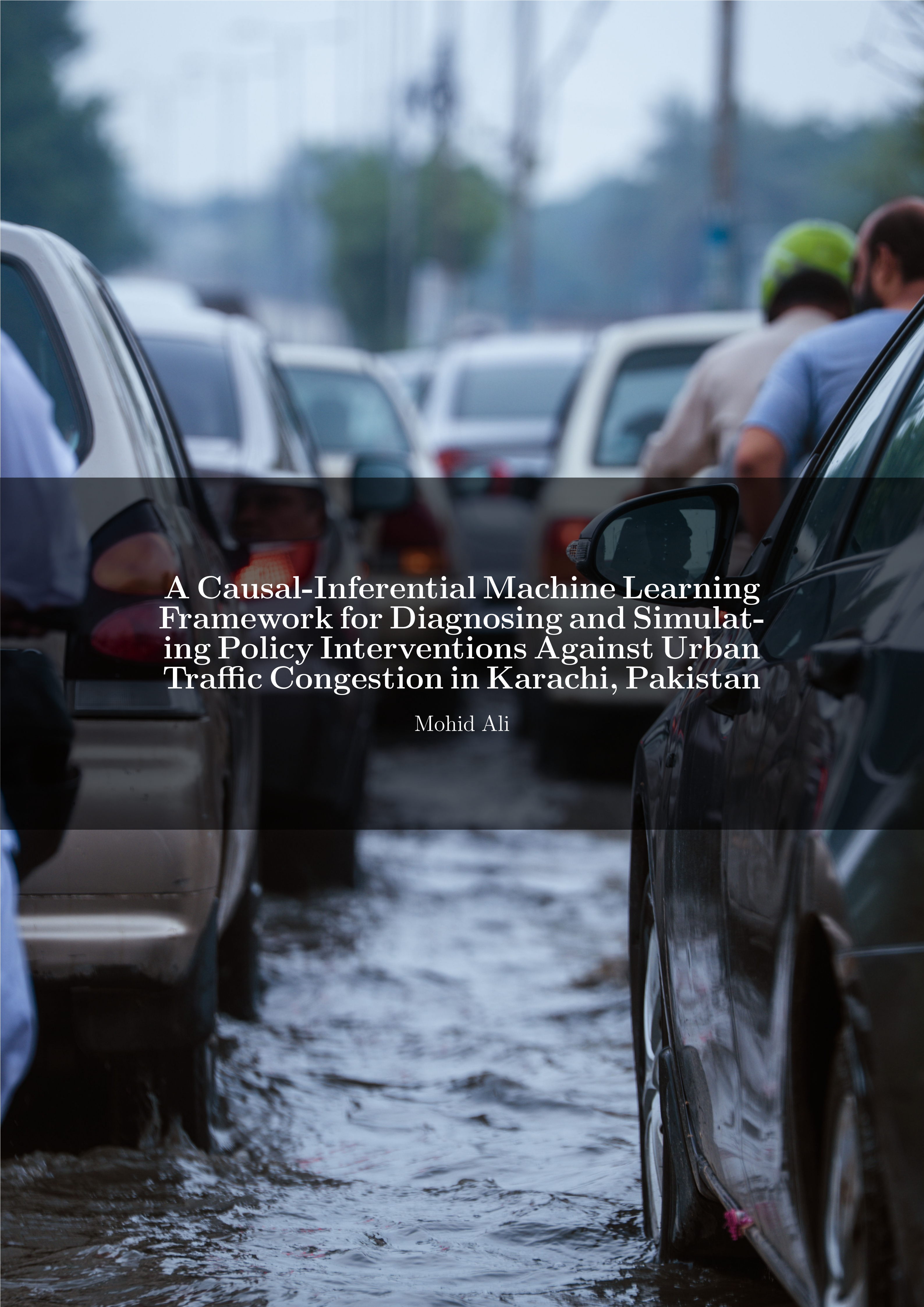

A Causal-Inferential Machine Learning Framework for Diagnosing and Simulating Policy Interventions Against Urban Traffic Congestion in Karachi, Pakistan

Mohid Ali

Abstract

Urban traffic congestion in Pakistan is almost always discussed in purely descriptive terms – too many cars, bad roads, no discipline – with very little quantitative work asking what actually drives it, or whether a proposed intervention would causally reduce it. This project builds an end-to-end congestion analysis framework for Karachi, combining a calibrated traffic model for 48 major road segments with three years (2022–2024) of real hourly weather data, a full OpenStreetMap road network, and an incident log, to produce a 1.26-million-row hourly dataset. Three predictive models are trained: an XGBoost classifier for severe-congestion detection, an LSTM for short-term per-segment forecasting, and a spatio-temporal graph neural network (ST-GCN) for joint multi-segment forecasting. A causal directed acyclic graph is estimated using ordinary least squares, propensity score matching, and double machine learning (DoWhy/EconML) to move beyond correlation and identify which road interventions actually reduce congestion. The central finding reverses a naive correlational read of the data: ordinary least squares suggests bus bays are associated with *more* congestion, but this is selection bias – Karachi built bus bays on its busiest corridors – and once road characteristics are controlled for, propensity score matching and double machine learning both show bus bays causally *reduce* congestion (DML ATE $= -0.021$, 95% CI $[-0.027, -0.016]$). Ten policy interventions are then ranked by simulated city-wide impact, with a full bus-bay rollout emerging as the single largest scalable lever (-8.6% mean congestion), ahead of congestion pricing, truck bans, and road widening. All of the above – live maps, forecasts, hotspot clusters, and the policy simulator – are exposed through an interactive Streamlit dashboard.

Contents

1 Introduction

Traffic congestion is one of the most visible costs of rapid, under-planned urbanization, wasting fuel, time, and productivity while disproportionately affecting the residents who can least afford the delay. Karachi, Pakistan's largest city, is a frequently cited example of a congested metropolis, yet almost every public discussion of *why* it is congested relies on anecdote, not data, and almost no existing work asks whether a specific policy – adding a bus bay, widening a road, pricing the CBD – would actually *cause* lower congestion, as opposed to merely correlating with it in whatever direction the existing infrastructure happens to point.

This project addresses that gap with a framework built around three questions: (1) where and when is congestion in Karachi worst, (2) what causally drives it, and (3) what happens if a specific intervention is applied city-wide. The framework combines supervised machine learning for prediction, causal inference for explanation, and a treatment-effect-driven simulator for policy comparison, all delivered through an interactive dashboard.

2 Study Area and Data

2.1 Study Area

The analysis covers 48 monitored segments across 11 major road corridors spanning Karachi's central business district, its main arterials, and its peripheral bypass and expressway network – from the dense, low-speed core of II Chundrigar Road to the elevated, high-speed Lyari Expressway.

Road / Corridor	Segments	Free-Flow Speed (km/h)	Character
II Chundrigar Rd	4	40	CBD core, most congested corridor
M.A. Jinnah Rd	5	50	Dense commercial corridor
Shahrae Faisal	6	60	Main arterial, moderate–heavy
University Rd	5	60	Moderate, education/commercial belt
Rashid Minhas Rd	4	50	Moderate
Korangi Rd	5	50	Moderate, industrial belt
Hub River Rd	4	65	Light–moderate
Clifton	2	50	Light, residential/coastal
Sea View	1	45	Light, coastal
Northern Bypass	6	70	Light, peripheral bypass
Lyari Expressway	6	80	Elevated expressway, least congested

Table 1: The 11 monitored road corridors and their 48 constituent segments.

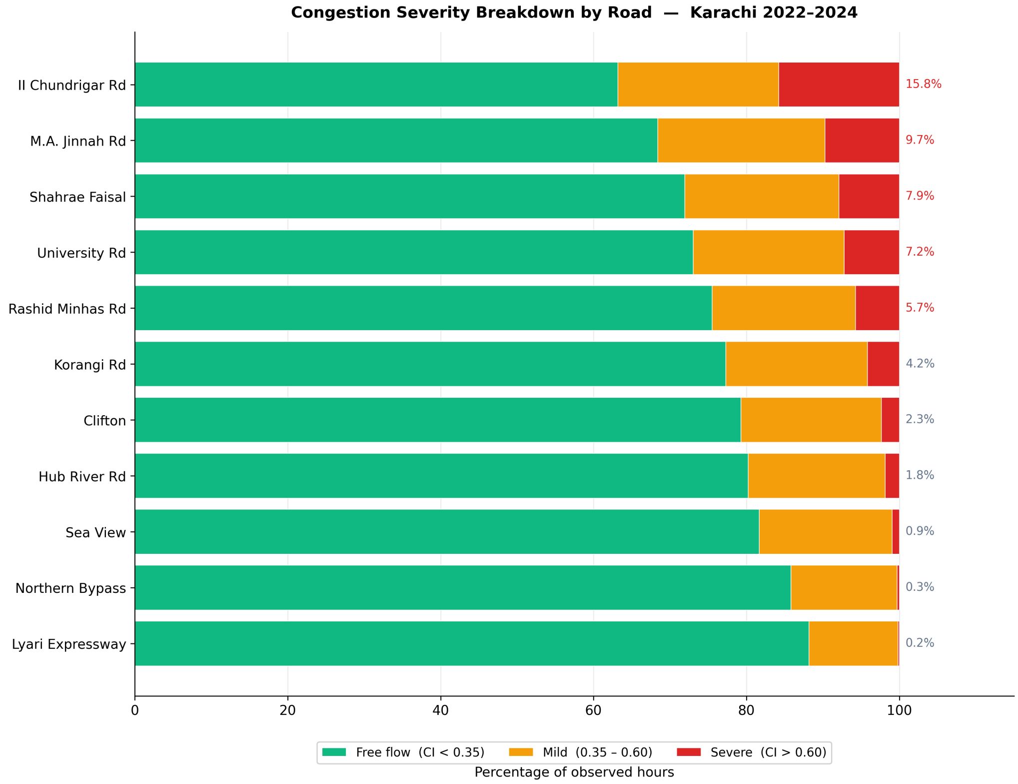

Figure 1: Congestion severity breakdown by corridor, 2022–2024. II Chundrigar Road spends 15.8% of observed hours severely congested – more than any other corridor and over 60% higher than the next-worst, M.A. Jinnah Road.

2.2 Data Sources

Data was assembled from four sources, each contributing a different layer of the analysis:

- **Calibrated Traffic Model**: hourly speed observations for 48 road segments, built from a time-of-day curve (Karachi rush hours 07:30–10:00 and 17:00–20:30) combined with real Open-Meteo weather, and adjustments for Ramadan, the school year, Friday Jumu'ah, weekends, rain, and extreme heat. Per-road congestion bias reflects known Karachi corridor behaviour (II Chundrigar worst, Lyari Expressway best). No traffic API currently covers Karachi at the resolution this project needed without a paid tier, so this calibrated model was used to generate a full three-year backfill (2022–2024), yielding 1,262,592 hourly observations. The provider layer is built to be swapped for a live API (TomTom or Google Maps) without changing any downstream code.

- **OpenStreetMap (via OSMnx)**: the full directed road network of Karachi – 9,453 nodes and 18,999 edges spanning 6,256 km – used for graph centrality analysis, isochrone reachability maps, and to match each of the 48 monitored segments to its nearest real road edge.

- **Open-Meteo**: 26,304 hourly weather records (2022–2024) – temperature, humidity, precipitation, wind speed, cloud cover, and pressure – plus derived flags for rain, heavy rain, fog, monsoon, and extreme heat.

- **Incident Log**: 728 accident, closure, protest, and VIP-movement records, calibrated to the real 2022–2024 weather record – accidents weighted by rainfall, flooding tied to heavy-rain hours, and protests and VIP movement weighted toward weekdays. (News scraping of Dawn.com and Geo News did not return enough usable articles for a standalone log, so this calibrated approach was used instead.)

2.3 Master Dataset Schema

All sources were merged into a single hourly, per-segment table spanning 2022–2024 across 48 road segments – 1,262,592 rows in total, 40 columns, with no missing values after cleaning. Key columns include:

- **congestion_index**: ratio of the speed deficit to free-flow speed (0 = free flow, 1 = gridlock)
- **speed_ratio**: observed speed divided by the segment's free-flow speed
- **speed_lag_1h / 2h / 3h**: the segment's own speed one, two, and three hours prior
- **speed_roll_mean/std_6h / 12h / 24h**: rolling mean and standard deviation of speed over 6, 12, and 24 hour windows
- **hour, day_of_week, month, is_weekend, is_peak_am, is_peak_pm, is_ramadan**: calendar and Karachi-specific time features
- **temperature_c, humidity_pct, precipitation_mm, wind_speed_kmh, cloud_cover_pct, pressure_hpa**: hourly weather variables joined from Open-Meteo
- **is_raining, is_heavy_rain, is_fog, is_monsoon, is_extreme_heat**: derived weather flags
- **incident_within_1km**: binary flag for a nearby recorded incident that hour
- **segment_id, road_name, lat, lon**: segment identity and geolocation

2.4 Data Preprocessing

Traffic timestamps were recorded in UTC while weather and incident timestamps were already in Karachi local time (UTC+5); this mismatch was resolved before any join by converting traffic timestamps to Asia/Karachi. Weather – recorded once per hour city-wide, not per segment – was joined to traffic on the truncated hourly timestamp, matching 100% of rows. Incident locations were mapped to the corresponding road corridor and joined on the same hourly key, flagging 1,331 rows with a nearby incident. The full pipeline ran through Apache Spark and was written to a partitioned Delta Lake warehouse (36 partitions) for downstream querying, and every processed table passed a validation pass confirming zero nulls and zero duplicate rows.

3 Methodology

3.1 Exploratory and Geospatial Analysis

Before modeling, the dataset was examined through univariate, temporal, and spatial exploratory analysis. Road-network betweenness centrality was computed across all 9,453 OSMnx nodes and combined with observed congestion into a bottleneck score; the single highest-scoring point sits on Shahrae Faisal (score 0.514). DBSCAN density-based clustering (eps $\approx$ 2.2 km, min_samples = 2) grouped the 48 segments into four geographic congestion hotspots plus six noise points, corresponding to the peripheral Northern Bypass segments. Four non-parametric Mann-Whitney U tests probed specific temporal hypotheses: congestion is significantly higher on weekdays than weekends (mean CI 0.254 vs. 0.164 – the largest effect found anywhere in the dataset) and during Ramadan ($p = 1.3 \times 10^{-43}$), the school-year-vs-summer difference is statistically significant but small ($p = 1.3 \times 10^{-11}$), and – counter-intuitively – the monsoon-vs-non-monsoon difference is *not* significant ($p = 1.00$): rain spikes congestion sharply during individual rain events, but Karachi's few rainy hours per year barely move the annual average. Isochrone reachability maps from five reference points (Saddar, DHA, SITE, Korangi, and the Airport) show every location loses 42–62% of its 45-minute reachable area during peak hours, with Korangi the worst hit and the Airport the least affected.

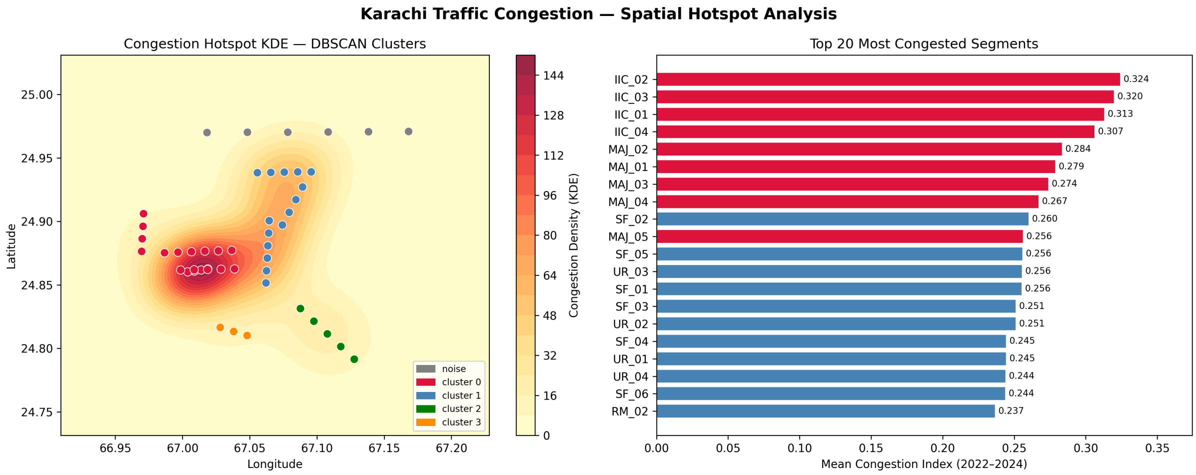

Figure 2: DBSCAN hotspot clustering (left) and the 20 most congested segments by mean congestion index (right). The densest cluster sits over the downtown core, matching the II Chundrigar and M.A. Jinnah corridors that dominate the ranking on the right.

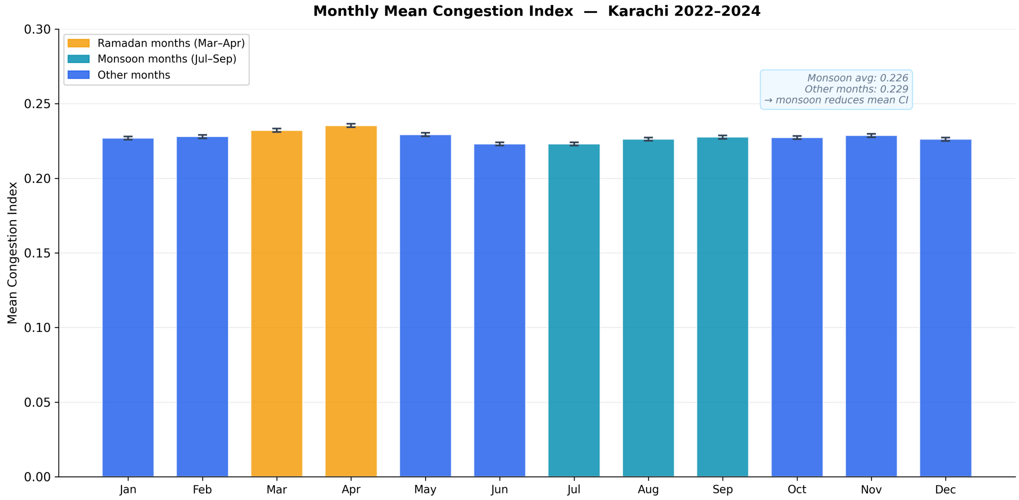

Figure 3: Mean congestion index by month, 2022–2024, with Ramadan and monsoon months highlighted. Ramadan (Mar–Apr) shows a small but consistent rise; the monsoon months (Jul–Sep) do not, despite rain sharply spiking congestion during individual events.

3.2 Predictive Modeling

Three complementary models were trained, each suited to a different question:

- **XGBoost (severe-congestion classification)**: a gradient-boosted tree classifier trained on 33 tabular features (excluding the raw speed and congestion-index columns that would leak the label) to flag hours where a segment's congestion index exceeds 0.6, tuned with 50 trials of Optuna hyperparameter optimization on a 50K-row subsample and interpreted with SHAP.
- **LSTM (per-segment forecasting)**: a two-layer recurrent network (hidden size 32) that consumes the past 24 hours of speed, weather, and time features for a single segment and forecasts its congestion index 1, 2, and 3 hours ahead.
- **ST-GCN (spatio-temporal forecasting)**: two graph-convolution layers over a 48-node segment graph, with edges weighted by inverse geographic distance, feeding a two-layer LSTM, forecasting all 48 segments' congestion jointly instead of one at a time.

3.3 Causal Inference

This is the central contribution of the project: descriptive and predictive traffic analysis is common, but almost none of it asks whether a specific road intervention actually *causes* lower congestion, as opposed to merely correlating with it. A structural causal model was defined as an eight-node directed acyclic graph (DAG): time-of-day and bus-bay presence drive vehicle volume, which mediates their effect on congestion, while rainfall, road width, signal presence, and incidents affect congestion directly. Using this graph, DoWhy confirmed all five candidate treatments (rainfall, has_bus_bay, signal_present, road_width, incident) are non-parametrically identifiable via the backdoor criterion. Each was estimated with backdoor-adjusted ordinary least squares on a 50K-row sample, and the bus-bay treatment – the one causal question with a clear, actionable policy lever – was additionally estimated with propensity score matching and EconML's LinearDML (double machine learning). Every OLS estimate was stress-tested with placebo permutation, random-common-cause, and data-subset refutation tests, and all five passed.

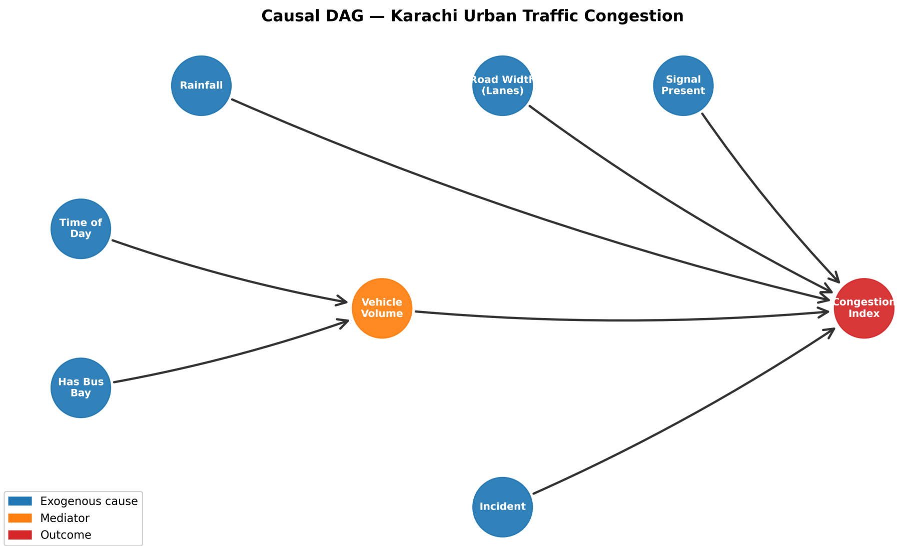

Figure 4: The structural causal graph used for effect estimation. Time-of-day and bus-bay presence act through the vehicle-volume mediator; rainfall, road width, signal presence, and incidents affect congestion directly.

3.4 Policy Simulation

The estimated treatment effects were used to simulate ten concrete interventions and rank them by projected city-wide impact on mean congestion. Supply-side interventions (bus-bay rollout, road widening, dedicated bus lanes) use the DML or OLS ATEs directly; demand-side interventions (congestion pricing, truck bans, staggered school times, park-and-ride) are modelled as a percentage reduction in vehicle volume, using reduction magnitudes drawn from international traffic-engineering literature, since Karachi has not implemented any of these measures and no local calibration data exists.

4 Evaluation Metrics

All predictive models used a strictly chronological train/validation/test split (2022-01-01 – 2023-08-31 / 2023-09-01 – 2024-02-29 / 2024-03-01 – 2024-12-31) instead of a random split, since shuffling hourly traffic data would let a model see the future during training and produce misleadingly optimistic results. XGBoost was evaluated with ROC-AUC and macro-F1 on the held-out test set; the LSTM and ST-GCN were evaluated with per-horizon Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) on the normalized congestion index.

Causal estimates cannot be evaluated against a held-out test set, since there is no ground-truth counterfactual to check against. Confidence instead rests on refutation testing – does the effect vanish under a placebo treatment, and does it survive the addition of an irrelevant random confounder – and, where possible, on triangulating multiple estimators against each other. The bus-bay treatment is the clearest example of why this matters: OLS alone would have produced the wrong policy conclusion, and only the combination of PSM and DML reveals the correct sign.

5 Results Summary

5.1 Predictive Model Performance

The XGBoost severe-congestion classifier reached a test ROC-AUC of 0.9994 and macro-F1 of 0.966, despite a heavily imbalanced target (only 5.23% of hours are severely congested). The LSTM forecaster reached a test MAE of 0.047 at one hour ahead, rising to 0.055 at two hours and 0.049 at three hours (0–1 scale), with validation and test metrics closely matched, indicating no overfitting. The ST-GCN spatio-temporal model underperformed the plain LSTM (test MAE 0.100 at one hour ahead), likely because the segment-adjacency graph is built on geographic proximity, not measured spillover between roads.

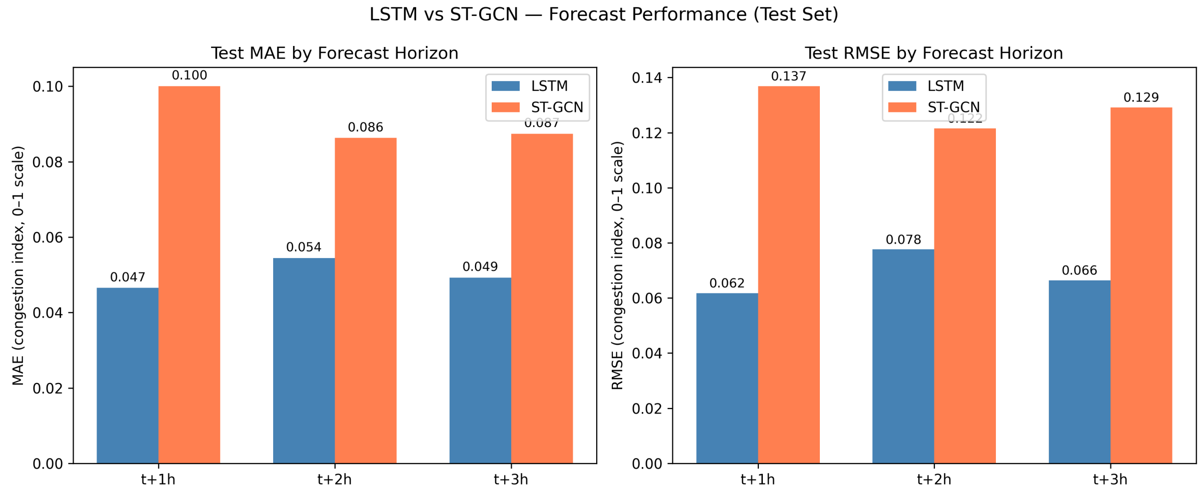

Figure 5: LSTM vs. ST-GCN test-set forecast error by horizon. The plain LSTM beats the graph-based model at every horizon, on both MAE and RMSE.

5.2 Causal Findings

The single most important finding in this project is a reversal, not a confirmation. Naive OLS regression finds that road segments with a bus bay have slightly *higher* average congestion (ATE = +0.007). Taken at face value, this would argue against building more bus bays. But this is selection bias: Karachi's bus bays were historically added to its already-busiest corridors, not placed at random. Once road characteristics are controlled for – first with propensity score matching (ATE = −0.038) and then with EconML's LinearDML (ATE = −0.021, 95% CI $[-0.027, -0.016]$) – the effect flips sign: bus bays causally *reduce* congestion. Signal presence (OLS ATE = +0.072) and incidents (OLS ATE = +0.156) show the expected direction and by far the largest magnitudes among the five treatments, while rainfall and road width show small, statistically detectable positive associations with congestion. All five OLS estimates passed both refutation tests.

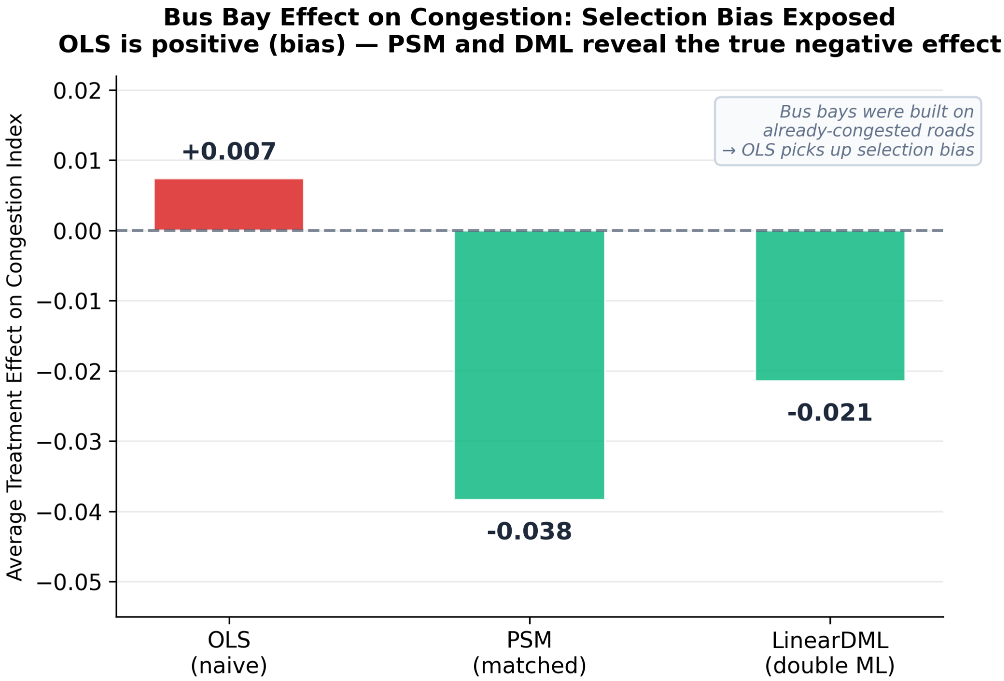

Figure 6: OLS, propensity score matching, and double machine learning estimates side by side, showing the bus-bay effect reversing sign once selection bias is controlled for.

5.3 Policy Simulation Results

Simulating all ten interventions shows a full bus-bay rollout as the single largest scalable lever available, reducing mean city-wide congestion by an estimated 8.6%, roughly double the next-largest option (a 5-road partial rollout at −4.9%). Demand-management interventions – congestion pricing and a peak-hour truck ban – cluster around −2.6%, and dedicated bus lanes on the two most congested arterials add a further −1.0% to −1.2%. Widening II Chundrigar Road by two lanes is the only intervention projected to *increase* congestion (+0.9%); this is the same selection-bias effect seen in the causal results, since Karachi's past road-widening decisions were made reactively on its most congested corridors, not pre-emptively.

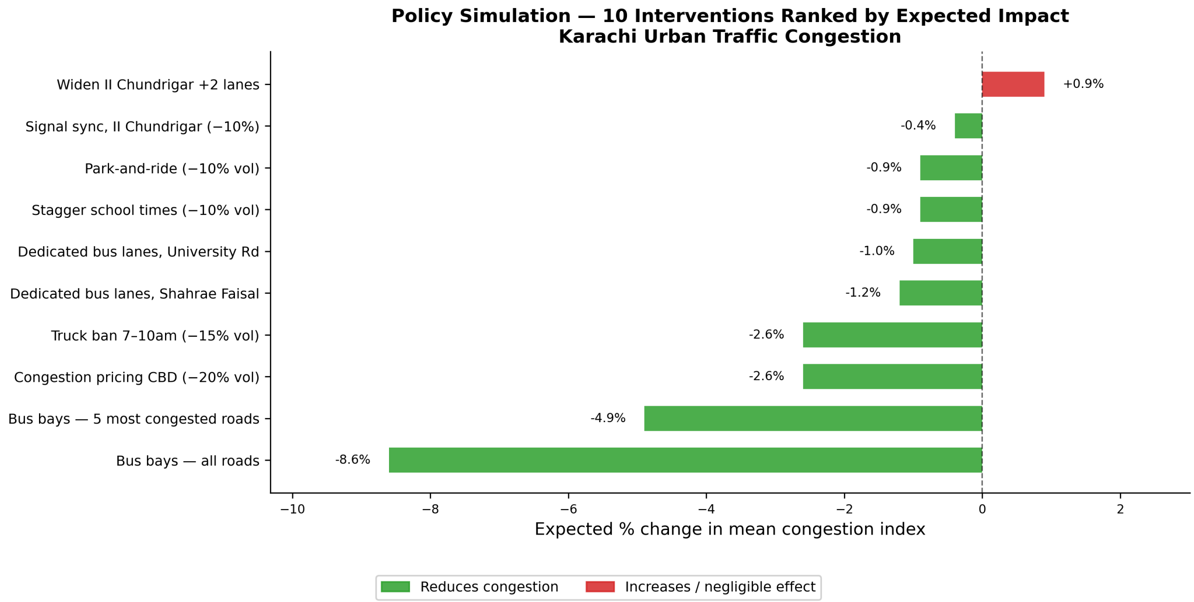

Figure 7: Ranked comparison of all ten simulated interventions by projected change in mean city-wide congestion.

6 Model Interpretability

Two complementary interpretability strategies were used, matched to each model's purpose. For XGBoost, SHAP (SHapley Additive exPlanations) values quantify how much each feature pushes a given severe-congestion prediction up or down, and consistently rank a segment's own recent speed history (lag and rolling-window features) and hour-of-day above weather and incident flags – congestion, in other words, is highly autocorrelated and time-of-day-driven, with external shocks acting as a secondary push. For the causal component, interpretability is built into the method itself: the DAG is an explicit, human-readable structural model, so the bus-bay reversal above can be traced to a specific, inspectable assumption – that road characteristics confound the raw bus-bay/congestion relationship – not treated as a black box. The LSTM and ST-GCN, by contrast, are used purely for forecasting and are evaluated on held-out accuracy, not interpreted feature-by-feature.

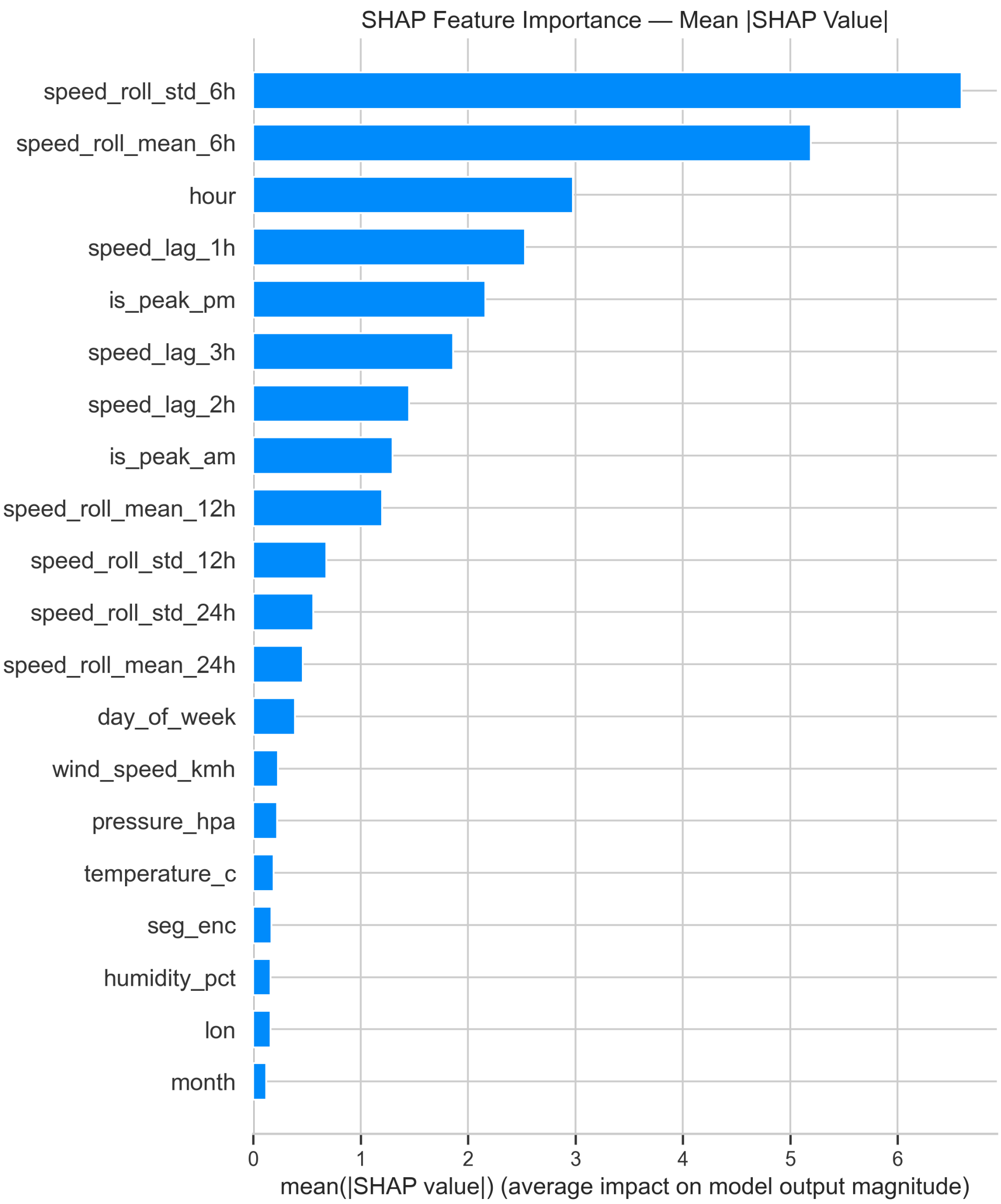

Figure 8: SHAP feature importance for the XGBoost severe-congestion classifier, ranking the strongest predictors.

7 Streamlit Dashboard

All of the above – the live data, the three predictive models, the causal findings, and the policy simulator – are exposed through a live, interactive dashboard built with Streamlit. The dashboard is deployed at `https://karachi-traffic-congestion-kdjanskdrnggc47gjsvpfm.streamlit.app/` and is publicly accessible. It is organized into four pages, described below.

7.1 Live Map

A Folium map of all 48 monitored segments, color-coded by their most recent predicted congestion index, giving an at-a-glance view of which corridors are currently worst affected, alongside headline city-wide statistics.

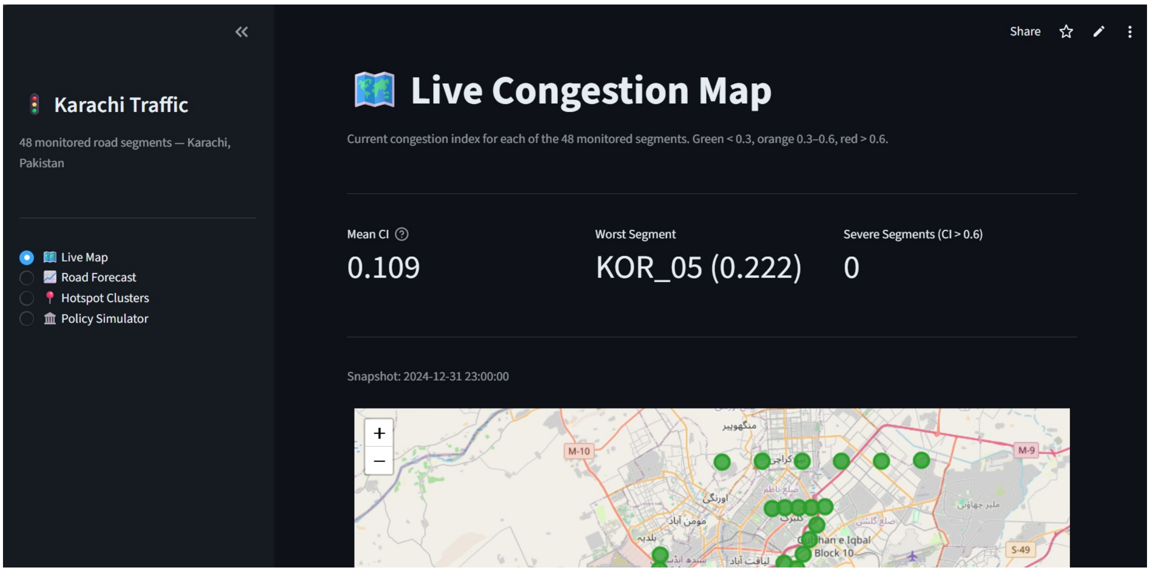

Figure 9: Live Map page, showing headline congestion statistics and the color-coded map snapshot across all 48 monitored segments.

7.2 Road Forecast

Selecting a road and segment runs the trained LSTM forward to produce a 3-hour congestion forecast, alongside a SHAP bar chart explaining which features are driving that segment's prediction.

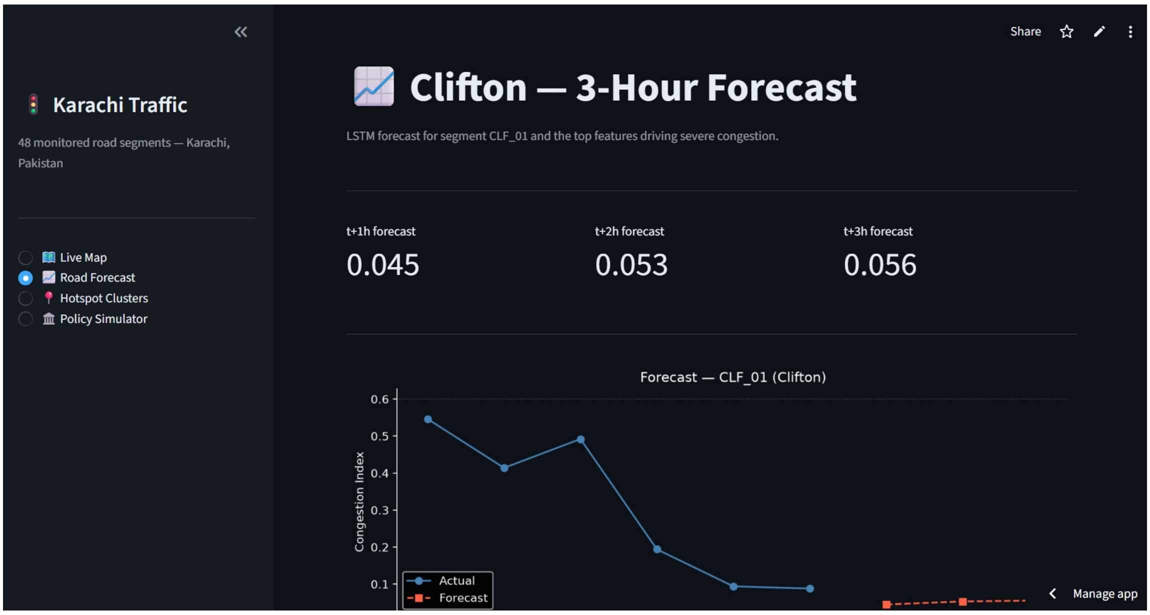

Figure 10: Road Forecast page, showing headline forecast statistics and the LSTM's 3-hour congestion forecast for a selected segment.

7.3 Hotspot Clusters

Displays the DBSCAN-derived hotspot clusters on a Folium map, alongside each cluster's hourly congestion profile and a ranked table of the worst individual segments.

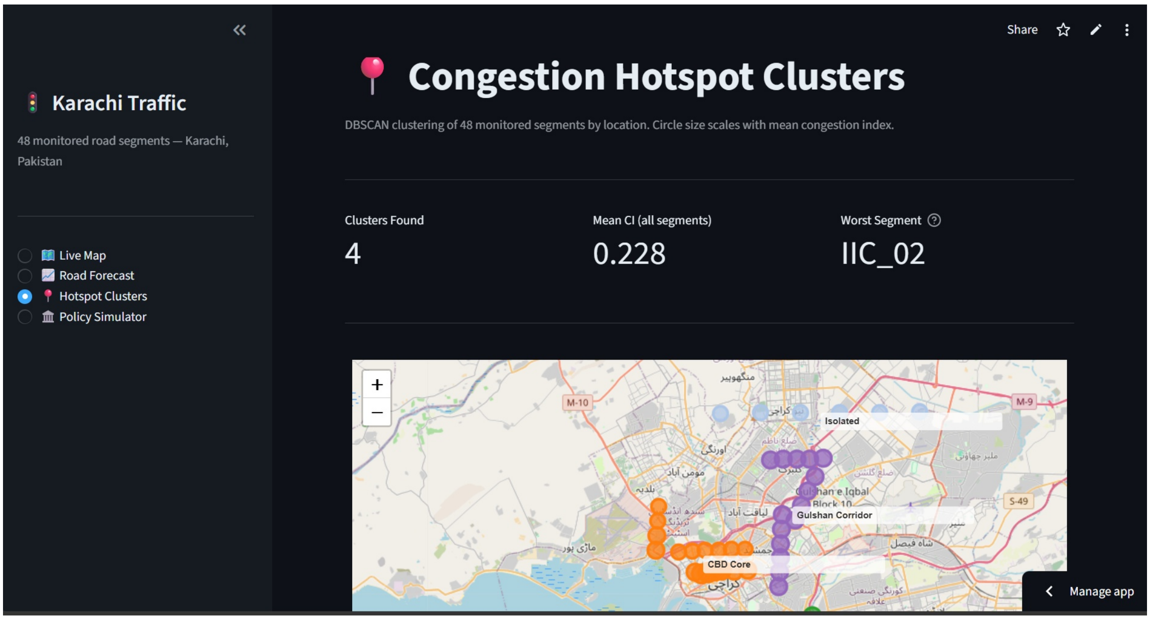

Figure 11: Hotspot Clusters page, showing cluster-level statistics and the DBSCAN cluster map across Karachi's road segments.

7.4 Policy Simulator

Lets the user see the single best-performing intervention at a glance, then pick any of the ten simulated interventions to see its ranked projected impact on city-wide congestion alongside a detail panel.

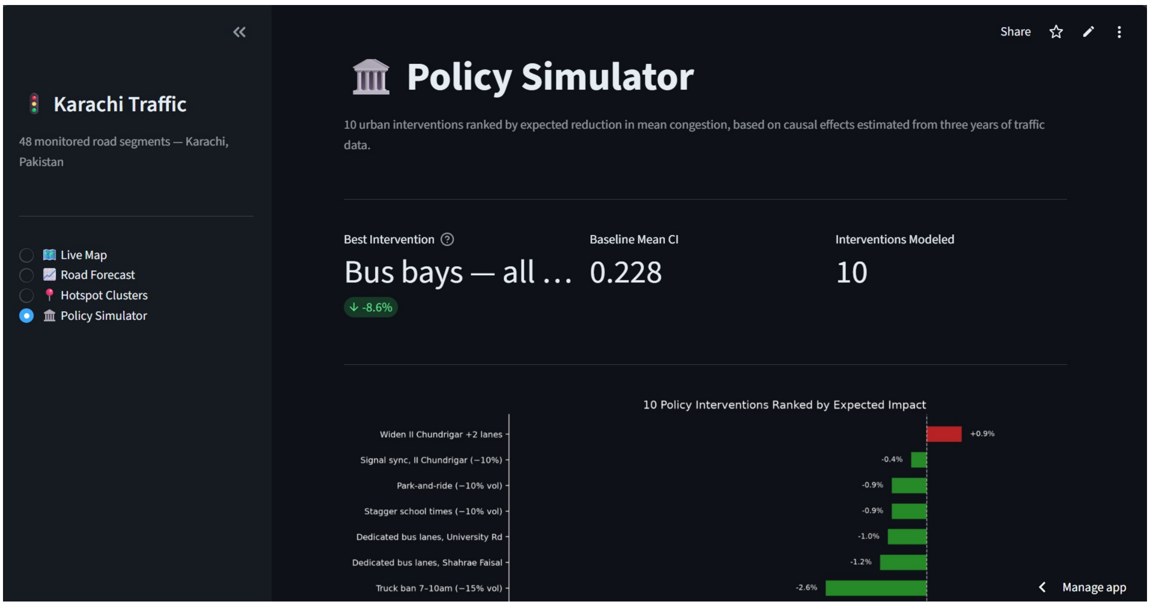

Figure 12: Policy Simulator page, showing the best-performing intervention and the ranked bar chart of all ten simulated interventions.

7.5 Accompanying Article

Alongside this technical report, the project's findings are presented in a long-form article for a general audience, published on Medium as *"Karachi's Traffic Is Not What You Think It Is"*. The article walks through the same core findings – which roads are actually the worst, the counter-intuitive monsoon result, the bus-bay selection-bias reversal, and the policy ranking – in accessible, non-technical language for a general audience, and links back to the live dashboard for readers who want to explore the results themselves.

8 Ethics and Limitations

This project is a research and educational tool, not an operational traffic-engineering instrument, and its policy ranking should not be read as a construction-ready recommendation without further local calibration.

First, some of the underlying data is synthetic, not directly measured. Traffic speeds come from a calibrated model built on a time-of-day curve, real Open-Meteo weather, and known Karachi corridor behaviour, since no traffic API currently covers Karachi at the resolution and cost this project required; the incident log was generated on the same principle, calibrated to real weather patterns, after automated scraping of Dawn.com and Geo News returned too few usable articles to build a reliable log. The pipeline is API-agnostic by design – swapping in TomTom or Google Maps data requires no change beyond the data-collection layer – and every result in this report should be read with that in mind.

Second, all causal estimates are derived from observational data, not a randomized experiment. Refutation testing and cross-estimator triangulation (OLS vs. PSM vs. DML) increase confidence that the bus-bay effect in particular is not spurious, but cannot rule out every unmeasured confounder, and the demand-side policy simulations (congestion pricing, truck bans, and similar) borrow reduction magnitudes from international literature, not Karachi-specific data.

Third, the ST-GCN's underperformance relative to the plain LSTM likely reflects that its segment-adjacency graph is built on geographic proximity, not measured spillover; a real sensor network could show a different ordering, since traffic genuinely propagates between adjacent roads.

9 Conclusion

This project delivers an end-to-end causal and predictive analysis of urban traffic congestion in Karachi, combining machine learning, causal inference, and an interactive simulation dashboard in a single framework. Its central finding – that a naive correlational read of the data would have led to the wrong conclusion about bus bays, and that causal methods reverse it – is a concrete demonstration of why causal inference, not just prediction, matters for transport policy. The resulting policy ranking gives a data-driven starting point: a city-wide bus-bay rollout is the single largest lever available among the options tested. Future work includes replacing the calibrated traffic layer with a live sensor or probe-vehicle feed as one becomes available, extending the incident log with verified reports, and testing the framework against other Pakistani cities.

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